

6 Cotransporter proteins are membrane proteins found in companion cells of phloem tissue.

Messenger RNA (mRNA) is the molecule in cells that carries genetic information in the DNA that codes for cotransporter proteins to the sites of protein synthesis in the cytoplasm.

(a) Complete Table 6.1 to compare the structure of a molecule of mRNA with the structure of a molecule of DNA.

Table 6.1

feature	mRNA	DNA
names of four bases		
name of pentose sugar present		
number of strands		

[3]

(b) Fig. 6.1 shows the events that occur in the nucleus of a companion cell in phloem tissue to synthesise molecules of mRNA.

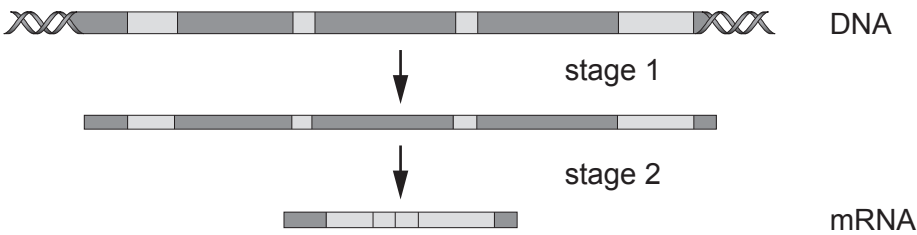


Fig. 6.1

(i) Name stage 1 shown in Fig. 6.1.

..... [1]

(ii) Describe what happens at stage 2, shown in Fig. 6.1, to shorten the length of the RNA molecule.

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..... [2]

- (c)** Cotransporter molecules are proteins produced in companion cells.

Fig. 6.2 shows what happens in the cytoplasm of a companion cell to a transfer RNA molecule before the cotransporter proteins can be produced.

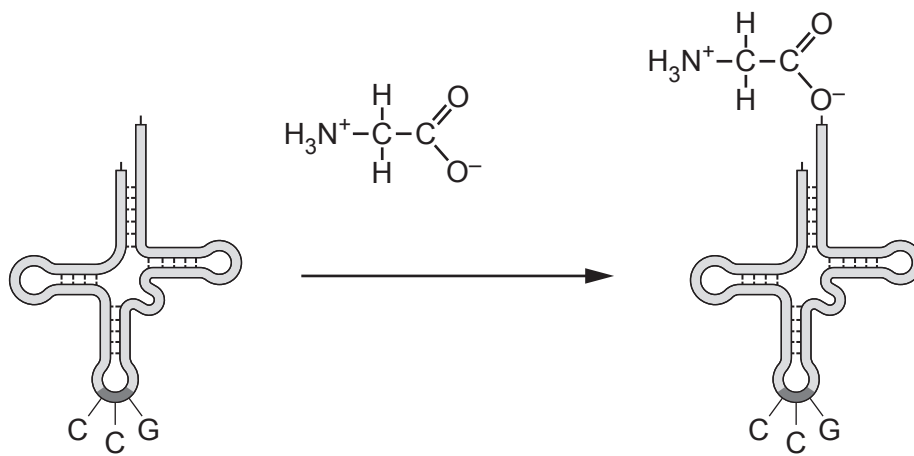


Fig. 6.2

- (i) Describe the role of the transfer RNA shown in Fig. 6.2 in the synthesis of a cotransporter protein.

[4]

(ii) Outline the role of cotransporter proteins in companion cells.

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..... [3]

[Total: 13]